

CSCI 5122

Neural Networks and Deep Learning for Data Science

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Machine Learning Engineer (60.3 / 100)	50.6 / 100

51

CURRENT AVG SCORE
across 6 job roles

89

POTENTIAL SCORE
with recommended additions

+54

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Explores neural networks and their application to real-world data science and AI problems. Covers neural network theory, algorithms, programming, and applications across various data formats.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	60.3	97.8	+37.5	HIGH
ML Engineer	58.7	89.3	+30.5	HIGH
Data Scientist	53.4	98.3	+44.9	HIGH
AI Engineer	52.3	83.3	+30.9	HIGH
Applied Scientist	45.6	100.0	+54.4	HIGH
MLOps Engineer	33.5	66.1	+32.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Deep Learning 2 of 6 roles

cnn

training

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 HIGH Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 MEDIUM Consider Deep Learning content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: cnn, training.

3 LOW Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

60.3 → 97.8 (+37.5 pts)

sets learning analytics business mobile design models machine

ML Engineer

58.7 → 89.3 (+30.5 pts)

cnn discovery learning machine systems models build impact

Data Scientist

53.4 → 98.3 (+44.9 pts)

sets analytics social build approaches

AI Engineer

52.3 → 83.3 (+30.9 pts)

following models learning build systems technology detection machine

Applied Scientist

45.6 → 100.0 (+54.4 pts)

discovery scientific research literature models learning systems industry

MLOps Engineer

33.5 → 66.1 (+32.6 pts)

detection systems learning machine image training models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.